

BONDS

Valuation for Financial Engineers
FRE 6103

David C. Shimko
david.shimko@nyu.edu

Outline of today's talk

- Theory
  - Idealized bonds
  - Variable but known interest rates
  - Spot and forward curves
  - Theories of the term structure
  - Pricing a bond for future delivery
  - Idealized floating rate notes (FRN)
  - Pricing when rates are stochastic
- U.S. Government Bonds
  - Composition
  - Details of bond types
  - The ZCB term structure
  - Bootstrapping
  - Inflation and government bonds



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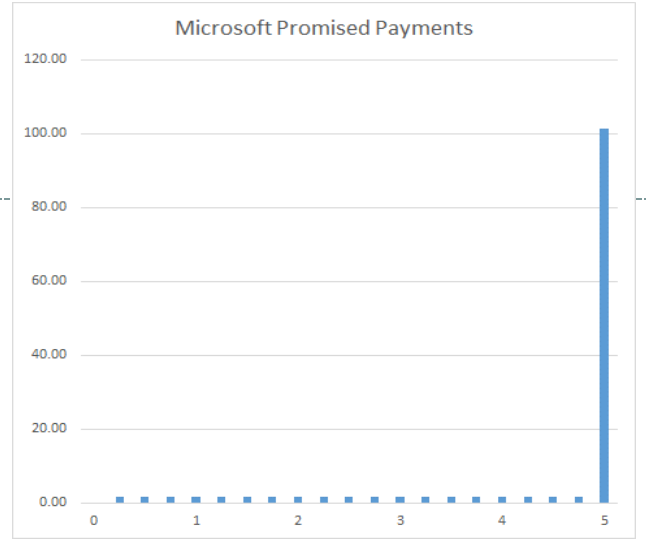
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Idealized bonds

- Idealized bonds match the Excel model
  - First payment is in one period
  - Equally spaced payments
  - Discount rate is constant
  - YTM (Bond equivalent yield) = Periodic rate × Periods/yr
  - Principal is $100 or $1000
- In this case, pricing is trivial
  - Excel provides price as a function of YTM or YTM as a function of price
    - See the PV and RATE function references

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Example



- If Microsoft issues a 6% coupon bond with a five year maturity, assuming $100 in principal value and quarterly coupons…
  - What is the coupon amount and when is it paid?
  - What is the value of the bond if the yield to maturity is 6%? What if the YTM rises to 7%?
  - What risks does the investor take when buying a Microsoft bond?
    - 1.
    - 2.
  - If the bond is priced at $108, what is its yield to maturity?

Note: The annual rate is converted to a quarterly rate by dividing by 4 (not compounding)

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Spot rates and forward rates

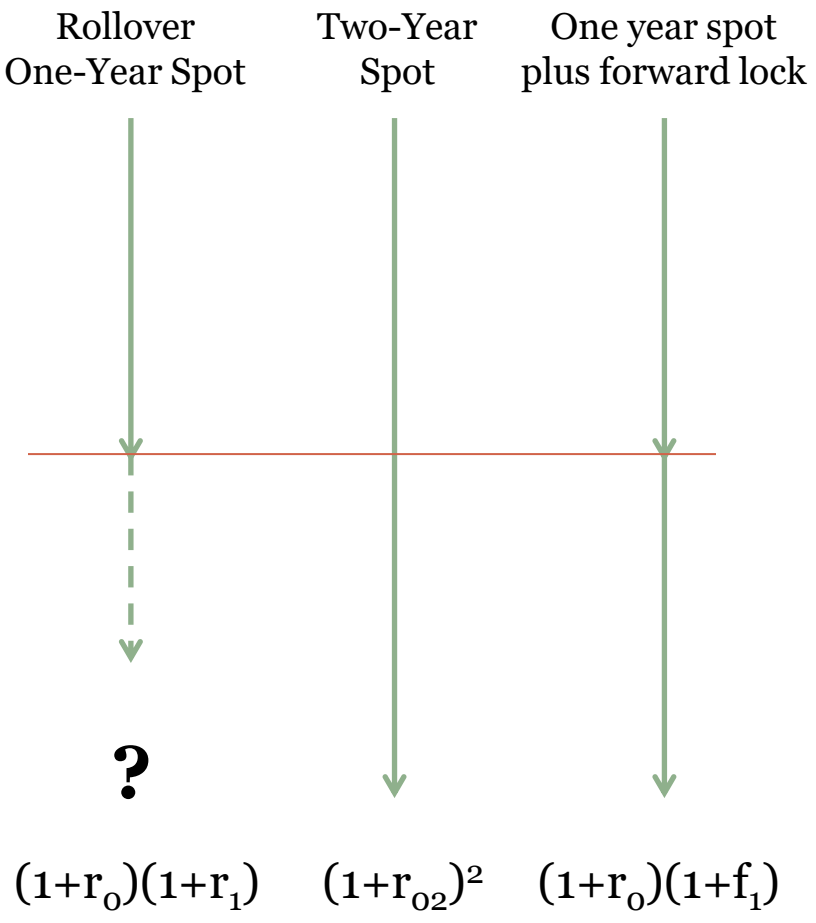
- The spot rate r_{0T} is the annual interest rate you lock in now (time 0) for the term of the debt (T)
- The forward rate (e.g.) f_1 is the implied return from year 1 to year 2
  - Obtained by buying the two-year bond and financing it by selling the one-year bond
- For continuous discounting, the spot rate is the simple arithmetic average of the one-year forward rates
  - As long as the interval between dates is constant
- Forward rates rise faster than spot when the term structure is upward-sloping

Time	Spot	Forward
1	2.0000%	2.0000%
2	2.2475%	2.4950%



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Three Ways to Invest $1 for Two years



Use this calculation to compute forward rates in discrete discounting models

The forward rate is determined implicitly by relative debt pricing:

$$(1+r_{02})^2 = (1+r_0)(1+f_1)$$

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Theories of the term structure

- If we knew what determined forward rates, we could determine spot rates
- There are four theories of the level of forward rates

1. Expectations Hypothesis $\mathbf{f = E(r_s)}$ REJECTED by data
   - Forward rates are expected future spot rates
2. Market Segmentation $\mathbf{f \neq E(r_s)}$ Untestable
   - The markets for short and long-term debt are different
   - Therefore rates are determined independently
3. Preferred Habitat $\mathbf{f > E(r_s)}$
   - Corporations prefer to issue long term debt
   - Investors prefer to invest in short term debt
   - Rates must adjust to induce corporations to borrow shorter term and investors to invest longer term
4. Liquidity Preference $\mathbf{f > E(r_s)}$
   - Investors value liquidity and low risk, so there is a built-in risk premium for longer term debt
   - Note that longer term bond prices are more sensitive to yield changes

CONSISTENT with data

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Yield curve movie

- Search for: Cara McDaniel Automated yield curve
- (There are many choices of movie)
- https://www.google.com/search?q=yield+curve+movie&rlz=1C1CHBF_enUS920US920&oq=yield+curve+movie&gs_lcrp=EgZjaHJvbWUyBggAEEUYOdIBCDMxOTlqMGo3qAIHsAIB&sourceid=chrome&ie=UTF-8#fpstate=ive&vld=cid:e1ab922d,vid:UH98TNzQmhQ,st:0
- As you watch, make note:
  - Is the curve generally upward or downward sloping?
  - How does the yield curve move? Parallel? Not parallel?
  - Which moves more, the short rate or the long rate?
  - Is the yield curve ever humped? (not monotonic)
  - What is the range of interest rates over the time period shown?

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How does theory explain upward slope?

- Well, ok, most of the time!
- Expectations hypothesis ✗
  - People generally expect rates to rise
- Market segmentation ?
  - Relative supply/demand for short and long-term debt
- Preferred habitat/Liquidity preference ☑
  - Even if people expect rates to fall, the term premium can make the curve slope upward
- General equation
  - Forward rate = Expected future spot rate + Term premium

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Exercises

- Question:
- Using the pricing model we developed, a client has asked you to deliver a one-year bond in 1 year, that pays $100 one year later.
  - This is called a *forward bond*
  - What is the fair price of that bond on delivery (using symbols)?
- Answer:
  - Price forward transactions by discounting with the forward discount rates
- Question:
- What is an equivalent strategy to holding a 2-year bond, using this forward transaction?
- Answer:
  - Buy the one-year bond that pays for the contracted bond at maturity, and enter the contract to purchase the bond in the second year

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The idealized floating rate note: GVE

- The FRN interest payment is different every period according to the prevailing (lagged) interest rate at the time
  - Most FRNs trade in the Eurodollar market and pay interest quarterly
- At time 0, the first payment of the FRN is known.
  - Pays principal (F) plus known interest at time 1.
  - This is called "payment in arrears"
  - MBE: $r_0 V_0 = (F - V_0) + r_0 F, \;\; so \; V_0 = F$
- The spot rate from time 1 to time 2 (r_1) is currently unknown, but can still be used in the equation
  - $r_1 V_1 = (F - V_1) + r_1 F \;\; or \;\; V_1 = F$
  - $r_0 V_0 = (V_1 - V_0) + r_0 F \;\; or \;\; V_0 = F$
- Therefore, pricing an (idealized) FRN is easy!

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Quiz questions

- How would you price an FRN between payment dates?

- What is the interest rate sensitivity of an FRN? (*Hint: Think of another fixed income instrument that has the same risk profile.*)

- Fill in the blanks:
  - "A fixed rate bond has _____ payments but _____ market value,
  - while an FRN has _____ payments but (approximately) _____ market value.

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A more difficult calculation

- How do you value a single FRN payment?
  - What is the current value of a cash flow of r_1F paid in one year, given that r_1 is unknown at time 0?
  - F can be a principal amount or any other constant multiplier
- Answer
  - This is a second use of forward rates (in addition to pricing forward bonds)
  - The value of the cash flow is risk-adjusted by replacing it with its forward equivalent (for the previous year, i.e. in arrears)
  - The forward equivalent is discounted at the risk-free rate
  - Hence, the value is $V_0 = f_1 F e^{-r_0}$
- Proof
  - Value all the coupons of an FRN and add them up.

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Sample FRN Valuation – 6 years, annual pmts

Floating Rate Note Valuation - Individual Payments

Principal 1000

Date	Spot	Rate		Forwards	CF equiv	PV
1	r01	5.00%		5.00%	50.00	47.62
2	r02	6.00%	f1	7.01%	70.10	62.38
3	r03	6.50%	f2	7.51%	75.07	62.15
4	r04	6.75%	f3	7.50%	75.04	57.78
5	r05	6.88%	f4	7.40%	74.02	53.07
6	r06	6.94%	f5	7.24%	1072.41	717.00
						1000.00

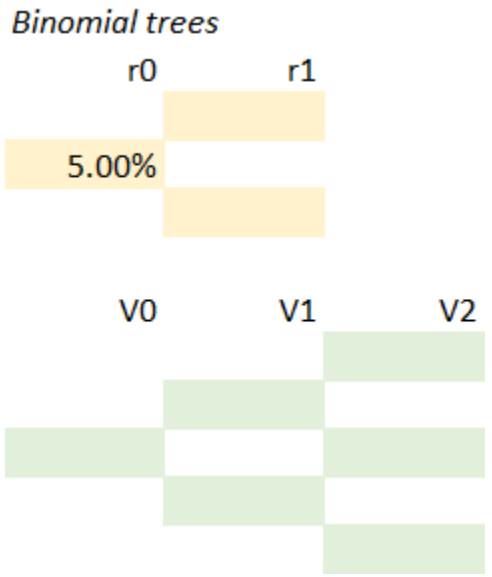
**Note: This slide uses discrete (not continuous) discounting*

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Valuing fixed-rate risk-free bonds with stochastic interest rates (MBE)

- Simplified model
  - Three periods: 0,1,2
  - Discount bonds pay $1 at time 1 or 2
  - One year spot rates r_0 and r_1
  - r_0 known, r_1 may go up or down 1σ
  - r_{02} is the two-year spot rate

- Valuation strategy
  - Start at time 2 and work backwards in time
  - V1 is known conditional on r1
  - V0 can be found from the MBE
    - Careful! Remember, the MBE refers to asset values not interest rates
    - Measure of risk (σ) = volatility of bond price
      - (not interest rate)
    - Price per unit risk (k) = liquidity risk premium



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Let's do a numerical valuation

Binomial Model: Two-period ZCB

Assumes two equally likely outcomes, up and down the same amount *(after drift or mean adjustment)*

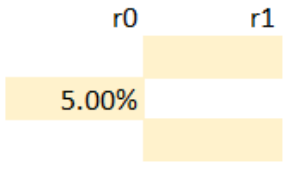
Parameters	Interp	Value
r0	Spot rate at time 0 for bond maturing at time 1	5.00%
μ	Spot rate expected change	1.00%
σ	Spot rate std dev from 0 to 1	2.00%
k	Cost of bond price risk	8.00%

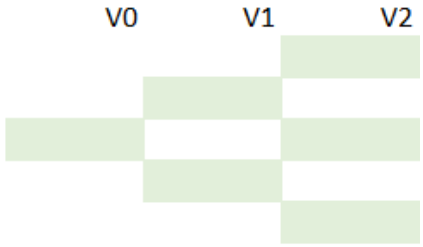
Outcomes	r1	V1
Up		
Down		

	E[V1]	SD.P[V1]	V0
Statistics			

r02	
f1	

Binomial trees





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Solution

Binomial Model: Two-period ZCB

Assumes two equally likely outcomes, up and down the same amount

Parameters	Interp	Value
r0	Spot rate at time 0 for bond maturing at time 1	5.00%
μ	Spot rate expected change	1.00%
σ	Spot rate std dev from 0 to 1	2.00%
k	Cost of bond price risk	8.00%

Outcomes	r1	V1
Up	8.00%	0.9259
Down	4.00%	0.9615

	E[V1]	StDev[V1]	V0
Statistics	0.9437	0.0178	0.8974

r02	0.0556
f1	0.0612

Binomial trees

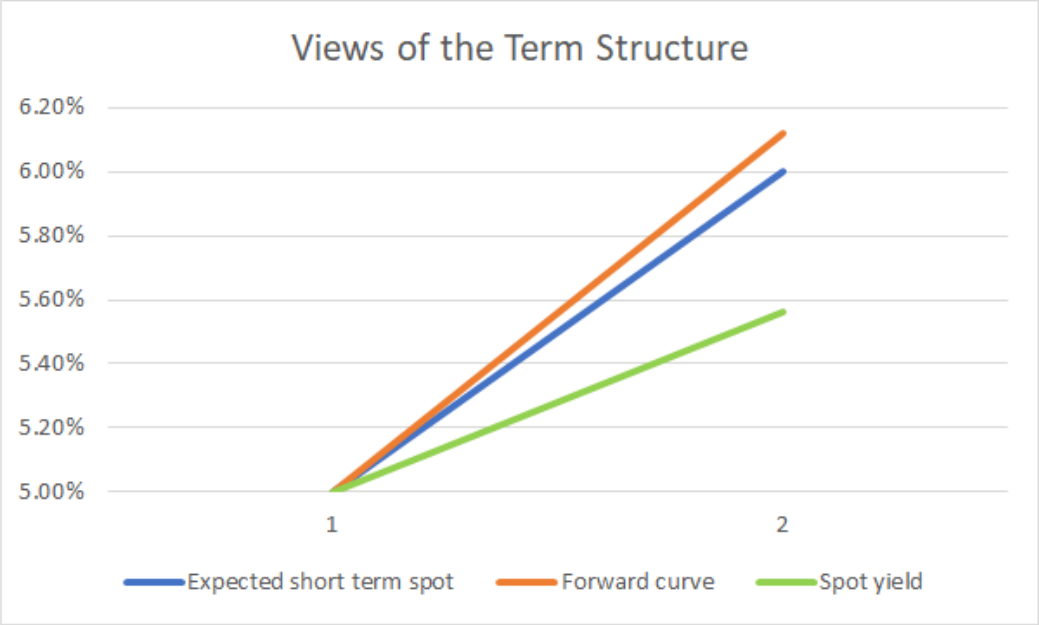
r0	r1
	8.00%
5.00%	
	4.00%

V0	V1	V2
		1
	0.9259	
0.8974		1
	0.9615	
		1

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Implications for the term structure views

Time	0	1
Expected	5.00%	6.00%
Forward	5.00%	6.12%
Spot	5.00%	5.56%



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Quiz

- Question
  - What is the risk-neutral bond valuation in the binomial model example?
- Answer
  - Discount the guaranteed payment at the _____ rates

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U.S. Government Bonds



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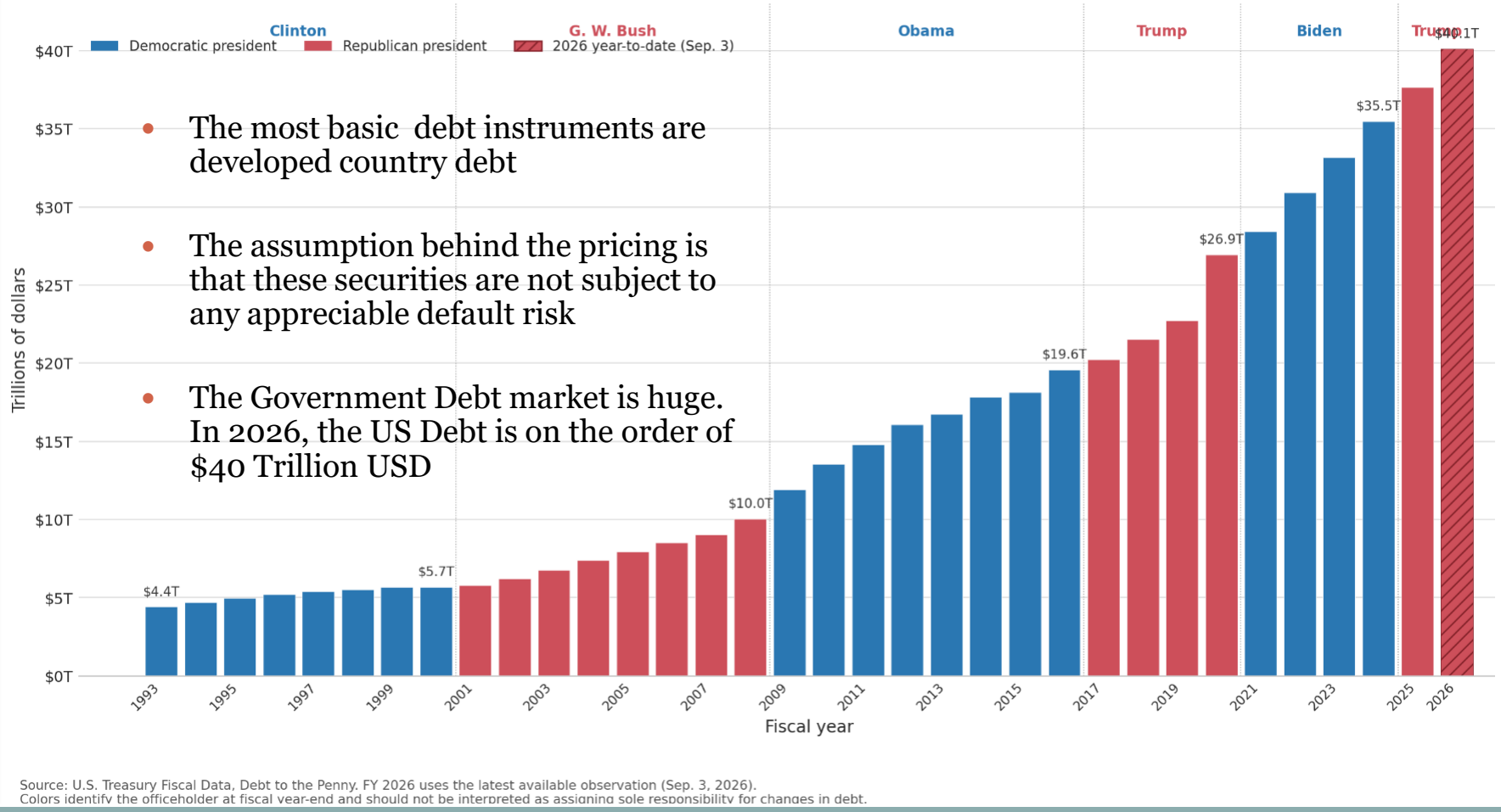
Fixed Income Terms

- Fixed Income: Any security that promises a fixed payment stream over time, or a payment based on a fixed formula
- Bond: A security that promises to make fixed or floating *coupon* payments at prescribed dates in the future until the *maturity date*, when the bond's *principal* is paid in full
  - The bond *issuer* is the borrower; the bond *investor* is the lender
  - U.S. government-issued bonds pay coupons semiannually (2x per year) and repay principal at maturity
  - U.S. corporate-issued bonds pay coupons quarterly (4x per year) and repay principal at maturity
  - U.S. mortgage bonds make payments monthly (12x per year) which include both principal and interest
    - Payment of principal prior to maturity is called *amortization*
- Loan: Similar payment structures to bonds, but there are important differences
  - Loans are not usually traded in the capital markets, e.g. bank loans
  - Loans usually have a lot more restrictions on the borrower
  - The interest rates on loans tend to be lower than bonds, but there is less flexibility for borrowers

- Floating rate bond: Payments are linked to changing interest rates, inflation rates or other indices
- Yield to maturity (YTM): Rate of return earned by a bond investor between now and the bond's maturity date (if the bond is held that long)
- Term structure: YTM for a given fixed income instrument for different maturities
- Zero coupon bond (ZCB): pays no coupons
  - Note that all fixed payment bonds can be imagined as the combination of a number of ZCBs
  - For this reason, we will focus on ZCB term structure and use continuous discounting
- Basis point: One bp = 0.01% = 0.0001 = "One bip"

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Gov’t Bonds – “Risk Free”



- The most basic debt instruments are developed country debt
- The assumption behind the pricing is that these securities are not subject to any appreciable default risk
- The Government Debt market is huge. In 2026, the US Debt is on the order of $40 Trillion USD

Source: U.S. Treasury Fiscal Data, Debt to the Penny. FY 2026 uses the latest available observation (Sep. 3, 2026).
Colors identify the officeholder at fiscal year-end and should not be interpreted as assigning sole responsibility for changes in debt.

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U.S. Debt Composition in Aug 2026

MONTHLY STATEMENT OF THE PUBLIC DEBT OF THE UNITED STATES

AUGUST 31, 2026

TABLE I -- SUMMARY OF TREASURY SECURITIES OUTSTANDING, AUGUST 31, 2026

(Millions of dollars)

	Amount Outstanding		Totals
	Debt Held By the Public	Intragovernmental Holdings	
Marketable:			
Bills	7,247,856	214	7,248,070
Notes	16,214,682	3,739	16,218,421
Bonds	5,512,729	12,556	5,525,284
Treasury Inflation-Protected Securities	2,151,957	703	2,152,660
Floating Rate Notes [19]	679,918	58	679,976
Federal Financing Bank [1]	0	3,591	3,591
Total Marketable [a]	**31,807,141**	**20,860** [2]	**31,828,001**
Nonmarketable:			
Domestic Series	10,709	0	10,709
State and Local Government Series	78,929	0	78,929
United States Savings Securities	146,943	0	146,943
Government Account Series	366,371	7,739,792	8,106,163
Other	4,895	0	4,895
Total Nonmarketable [b]	**607,848**	**7,739,792**	**8,347,640**
Total Public Debt Outstanding	**32,414,989**	**7,760,652**	**40,175,641**

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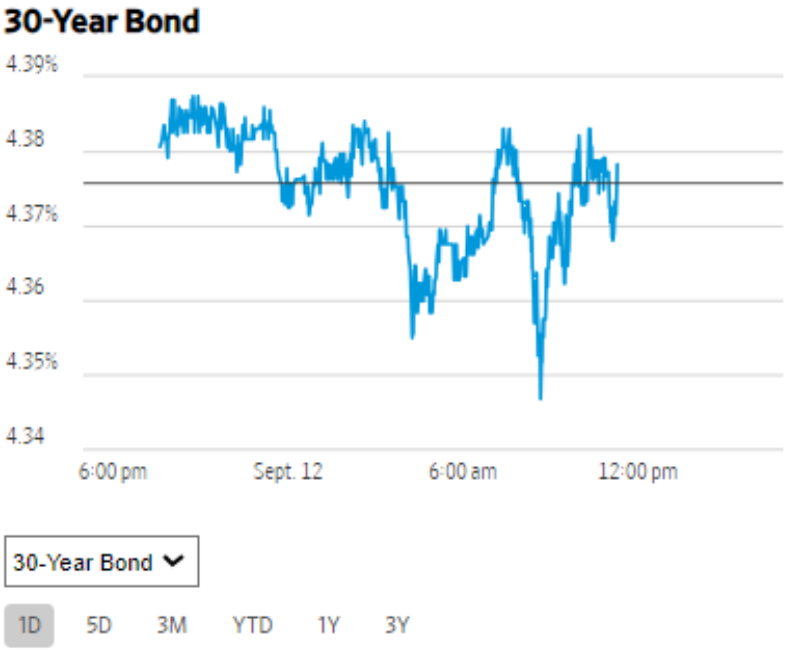
Sample Treasury bond quotes

WSJ, as of 9/12/23

U.S. Treasurys 11 17 AM EDT 9/12/23

	COUPON (%)	PRICE CHG	YIELD (%)	YIELD CHG
30-Year Bond	4.125	0/32	4.378	0.002
10-Year Note	3.875	0/32	4.287	-0.003
7-Year Note	4.125	0/32	4.376	-0.003
5-Year Note	4.375	0/32	4.431	0.011
3-Year Note	4.625	-19/32	4.693	0.030
2-Year Note	5	-22/32	5.020	0.025
1-Year Bill	0	0/32	5.417	0.005
6-Month Bill	0	0/32	5.540	0.008
3-Month Bill	0	0/32	5.439	-0.006
1-Month Bill	0	0/32	5.387	-0.006

View Treasury Quotes Page



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Problems with T-Bond YTM

- The payment pattern is specific to U.S. government bonds
- Other pricing objectives
  - Risk-free component of corporate bonds
  - OTC transactions with corporate customers
    - Long-term cash flow planning
  - Pricing derivatives
- We would like to know the discount rate for *every single bond payment*, not just the bond as a whole

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Why is the ZCB yield curve important?

- Any other fixed rate bond can be constructed and therefore priced from ZCBs
- Easy market references for risk-free ZCBs exist in the U.S. and most developed economies
  - T-Bill (treasury bill)
  - U.S. Government STRIPS
- We'd like to use continuous discounting to avoid any ambiguity

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What is a Treasury Bill?

- The T-bill is a ZCB issued by the U.S. government
- Primary market
  - Sold in weekly auctions to *primary dealers or direct to investors on Treasury.direct*
- Secondary market
  - Identified by CUSIP numbers
  - Re-sold in over-the-counter (OTC) transactions to other investors

- Face value $10,000
- Maturity less than one year
- Considered a risk-free investment

- Use this sample T-Bill price
  - Today's date 9/11/23
  - T-bill maturity date 10/24/23
  - Days to maturity = 43
  - Asked yield (NEXT PAGE) = 5.265%

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T-Bill Quotation (9/11/23)

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MarketData > Bonds & Rates > U.S. Treasury Quotes

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U.S. Treasury Quotes Monday, September 11, 2023

Treasury Notes & Bonds | Treasury Bills

Treasury bill bid and ask data are representative over-the-counter quotations as of 3pm Eastern time quoted as a discount to face value. Treasury bill yields are to maturity and based on the asked quote.

MATURITY	BID	ASKED	CHG	ASKED YIELD
9/14/2023	5.183	5.173	-0.048	5.261
9/19/2023	5.258	5.248	+0.013	5.340
9/21/2023	5.253	5.243	unch.	5.337
9/26/2023	5.273	5.263	+0.013	5.361
9/28/2023	5.275	5.265	+0.010	5.3653
10/3/2023	5.270	5.260	+0.002	5.364
10/5/2023	5.273	5.263	+0.003	5.354
10/10/2023	5.285	5.275	+0.018	5.385
10/12/2023	5.283	5.273	+0.005	5.384
10/17/2023	5.278	5.268	+0.013	5.383
10/19/2023	5.285	5.275	unch.	5.392
10/24/2023	5.275	5.265	+0.003	5.386

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T-Bill Conversions

- T-Bill quote to T-Bill price – "A yield is not necessarily a yield."
- T-Bill price to ZCB continuously compounded yield – helpful for comparisons to other yields
- Use this sample T-Bill price
  - Today's date 9/11/23
  - T-bill maturity date 10/24/23
  - Days to maturity = 43
  - Asked yield = 5.265%
- What is the price of the bill and the continuously compounded yield?

$$\text{Tbill price} = 10{,}000\left(1 - \text{Quoted yield} \times \frac{Days}{360}\right)$$

$$r = -\frac{\ln\left(\frac{Price}{10{,}000}\right)}{\left(\frac{Days}{365}\right)}$$

Continuously compounded yield

43 days; $9937.11; 5.355%

Why is the quotation so awkward?

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STRIPs

What are they?

- Allow investors to hold interest and principal components as stand alone securities

How quoted?

- $100 face value

$$EAY = \left(\frac{100}{Price}\right)^{365/days} - 1$$

Convert to continuous ZCB?

$$r = \ln(1 + EAY)$$

Verify the yield calculation for the indicated STRIP

2161 days
EAY = 1.89%
r = 1.88%

U.S. Treasury Strips

Wednesday, September 13, 2017

U.S. zero-coupon STRIPS allow investors to hold the interest and principal components of eligible Treasury notes and bonds as separate securities. STRIPS offer no interest payment; investors receive payment only at maturity. Quotes are as of 3 p.m. Eastern time based on transactions of $1 million or more. Yields calculated on the ask quote.

Maturity	Bid	Asked	Chg	Asked yield
Treasury Bond, Stripped Principal				
2018 May 15	99.242	99.249	0.001	1.13
2018 Nov 15	98.662	98.673	-0.017	1.15
2019 Feb 15	98.331	98.345	-0.021	1.18
2023 Feb 15	90.499	90.548	-0.114	1.84
2023 Aug 15	89.429	89.482	-0.115	1.89
2024 Nov 15	86.757	86.819	-0.104	1.98
2025 Feb 15	86.180	86.243	-0.158	2.01

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Bootstrapping T-Bills and STRIPs

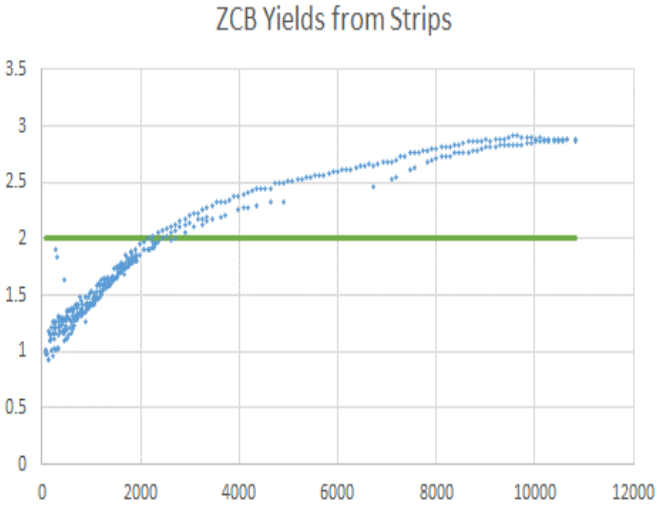
- Many different ways to estimate a full ZCB yield curve.
- In this case, imagine a flexible functional form, and use regression to fit the curve

Example:

$$y(t) = a_1 + a_2 t + a_3 t^2 + a_4 e^{a_5 t} + a_6 \ln(t - a_7) + \varepsilon$$

- Shown (green): a_1=2, else =0
- Objective: Minimize sum of squared deviations between model and actual yields.

As of 9/14/17	42992	243				
	Bid	Ask	Chg	Ask yield	Date text	Datevalue
2018 May 15	99.225	99.231	-0.017	1.16	May 15, 2018	43235
2018 Nov 15	98.648	98.659	-0.014	1.16	Nov 15, 2018	43419
2019 Feb 15	98.313	98.327	-0.018	1.2	Feb 15, 2019	43511
2019 Aug 15	97.478	97.496	-0.022	1.33	Aug 15, 2019	43692
2020 Feb 15	96.783	96.806	-0.023	1.35	Feb 15, 2020	43876
2020 Feb 29	96.645	96.668	-0.031	1.38	Feb 29, 2020	43890
2020 May 15	96.33	96.355	-0.033	1.4	May 15, 2020	43966
2020 Aug 15						44058
2021 Feb 15						44242
2021 May 15						44331
2021 Aug 15						44423
2021 Nov 15						44515
2022 Aug 15						44788
2022 Nov 15						44880
2023 Feb 15						44972
2023 Aug 15						45153
2024 Nov 15						45611
2025 Feb 15						45703
2025 Aug 15						45884
2026 Feb 15						46068
2026 Aug 15						46249
2026 Nov 15	82.085	82.159	-0.004	2.16	Nov 15, 2026	46341
2027 Feb 15	81.513	81.589	-0.008	2.17	Feb 15, 2027	46433
2027 Aug 15	80.488	80.567	0.007	2.19	Aug 15, 2027	46614
2027 Nov 15	79.865	79.945	unch.	2.21	Nov 15, 2027	46706



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T-Bond quotes – 9/2/20

- https://www.wsj.com/market-data/bonds/treasuries
- Treasury note and bond data are representative over-the-counter quotations as of 3pm Eastern time. For notes and bonds callable prior to maturity, yields are computed to the earliest call date for issues quoted above par and to the maturity date for issues below par.

MATURITY	COUPON	BID	ASKED	CHG	ASKED YIELD
9/15/2020	1.375	100.01	100.014	-0.004	-0.0621
9/30/2020	1.375	100.026	100.032	-0.006	-0.0017
9/30/2020	2	100.044	100.05	-0.006	-0.1168
9/30/2020	2.75	100.056	100.062	-0.01	0.101
10/15/2020	1.625	100.052	100.056	-0.004	0.059
10/31/2020	1.375	100.06	100.064	-0.006	0.086
10/31/2020	1.75	100.08	100.084	-0.006	0.064
10/31/2020	2.875	100.134	100.14	-0.01	0.098
11/15/2020	1.75	100.102	100.106	-0.008	0.056
11/15/2020	2.625	100.156	100.162	-0.008	0.064
11/30/2020	1.625	100.112	100.116	-0.008	0.097
11/30/2020	2	100.142	100.146	-0.008	0.082
11/30/2020	2.75	100.202	100.206	-0.008	0.052
12/15/2020	1.875	100.162	100.166	-0.01	0.015
12/31/2020	1.75	100.166	100.172	-0.01	0.082
12/31/2020	2.375	100.232	100.236	-0.008	0.079
12/31/2020	2.5	100.242	100.246	-0.01	0.107
1/15/2021	2	100.216	100.222	-0.014	0.09
1/31/2021	1.375	100.16	100.164	-0.01	0.109
1/31/2021	2.125	100.26	100.264	-0.012	0.092

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Bootstrapping from Bonds

- Bonds provide a deeper market than STRIPs, so banks prefer to bootstrap the term structure from bonds
- There are many ways to bootstrap bonds
  - Use the smooth function approach demonstrated for STRIPs 
  - Use a regression approach.
    - Each bond value is the sumproduct of its payment stream with its discount factors (i.e solve for $d(t)=e^{-rt}$ instead of r).
    - Find the discount factors that provide the best fit
      - This may require inversion of a large matrix, not suitable for Excel
  - Interpolation methods
  - Cubic splines
  - Maximizing smoothness

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Observations on T-Bond quotes

- Prices are “clean prices” and do not include accrued interest
  - See the ACCRINT function in Excel
- There are a relatively small number of dates when U.S. government bonds make coupon payments
  - We only need to include interest rates for those dates
- We’d like to find continuous discount factors for each of these payment dates
- These form the basis of all risk-free valuation

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MINI-PROJECT #2: ZCB Yield Curve

- Find a partner (one other person)
  - We will match you in class if you don't find someone!
- You may use Python or Excel
- For the most recent trading day, download all the US Bond prices from the Wall Street Journal or Bloomberg.
  - This has been done for you
- Develop a detailed automated algorithm to determine the *continuously compounded* discount rate for each bond payment date (coupons and principal).
  - In the form of a function (this may require some manual experimentation)
  - In the form of a graph
- Write up your methodology (less than one page), and show your results in a table and graph.

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What is inflation?

- What it means for spending power
- What it means for fixed rate liabilities
- **Nominal and real interest rates**
- Inflation equations: The *Fisher Equation*
  - Discrete time discounting
    - $\rho = \frac{r-i}{1+i}$
  - Continuous discounting (also used as a discrete time approximation)
    - $\rho = r - i$
  - *Have we seen this before somewhere?*

Key valuation conclusion:

Always discount _____ cash flows at _____ rates,

and _____ cash flows at _____ rates.

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The Consumer Price Index (CPI)



Data extracted on: Apr 30, 2019 (10:42:21 AM)

CPI-All Urban Consumers (Current Series)

Series Title	:	All items in U.S. city average, all urban consumers, not seasonally adjusted
Series ID	:	CUUR0000SA0
Seasonality	:	Not Seasonally Adjusted
Survey Name	:	CPI-All Urban Consumers (Current Series)
Measure Data Type	:	All items
Area	:	U.S. city average
Item	:	All items

Latest Observation:
March 2019
254.202

Jan 2017 - Mar 2019
Minimum Value: January 2017
242.839

Maximum Value: March 2019
254.202

Data Availability:
1913 - 2019

.XLSX .CSV

Year	Period	Label	Value
2018	M02	2018 Feb	248.991
2018	M03	2018 Mar	249.554
2018	M04	2018 Apr	250.546
2018	M05	2018 May	251.588
2018	M06	2018 Jun	251.989
2018	M07	2018 Jul	252.006
2018	M08	2018 Aug	252.146
2018	M09	2018 Sep	252.439
2018	M10	2018 Oct	252.885
2018	M11	2018 Nov	252.038
2018	M12	2018 Dec	251.233
2019	M01	2019 Jan	251.712
2019	M02	2019 Feb	252.776
2019	M03	2019 Mar	254.202

CPI is the price of a standardized "basket" of consumer goods, meant to represent changes in overall price levels.

What was the rate of U.S. inflation realized over the last year, according to the CPI?

ANSWER:
=(254.202/249.554)-1
= 1.86%

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Cost of living adjustments (COLA)

- Mingruo has just been offered an employment contract with Moody's Investor Services. The details are:
  - 5 year contract (assume end of year payments)
  - Current salary reference: $100,000 U.S. per year
  - Current CPI reference (3/19): 254.202
  - Future salary per year = Future CPI × $100,000 / 254.202
- How would you find the expected present value of these cash flows in the marketplace?
  - That is, using the market data provided in this presentation?

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Can we find the market's expected inflation?

Features of TIPS

As of 9/11/23

- Principal and interest calculations
  - Principal is increased/decreased with changes in CPI inflation; interest payments are tied to principal
- Taxation of capital gain
  - Increase of principal subject to federal tax, but exempt from state and local taxes
- Embedded option
  - Final Principal payments is at least par → insulated from deflation
- Implied inflation now?
  - Implied inflation ≈ Risk-free rate – rate on TIPS

MATURITY	COUPON	BID	ASKED	CHG	YIELD*	D PRINCIPAL
2024 Jan 15	0.625	98.28	98.3	2	3.848	1308
2024 Apr 15	0.5	98.02	98.04	1	3.494	1210
2024 Jul 15	0.125	97.18	97.2	1	3.011	1285
2024 Oct 15	0.125	96.28	96.3	1	2.986	1190
2025 Jan 15	0.25	96.1	96.12	1	3.038	1289
2025 Jan 15	2.375	98.32	99.02	1	3.099	1619
2025 Apr 15	0.125	95.18	95.2	1	2.967	1182
2025 Jul 15	0.375	95.3	96	1	2.642	1287
2025 Oct 15	0.125	95.02	95.04	1	2.542	1176
2026 Jan 15	0.625	95.14	95.16	2	2.62	1284
2026 Jan 15	2	98.14	98.16	1	2.658	1538
2026 Apr 15	0.125	93.27	93.29	1	2.569	1164
2026 Jul 15	0.125	93.28	93.3	1	2.338	1273
2026 Oct 15	0.125	93.16	93.18	1	2.292	1117
2027 Jan 15	0.375	93.18	93.21	unch.	2.362	1263
2027 Jan 15	2.375	99.29	99.32	1	2.377	1514
2027 Apr 15	0.125	92.12	92.16	unch.	2.319	1081
2027 Jul 15	0.375	93.13	93.16	unch.	2.145	1248
2027 Oct 15	1.625	97.29	98	unch.	2.131	1030
2028 Jan 15	0.5	93	93.04	unch.	2.172	1237
2028 Jan 15	1.75	98.03	98.06	unch.	2.19	1457
2028 Apr 15	3.625	106.08	106.11	unch.	2.164	1887
2028 Apr 15	1.25	96.01	96.04	unch.	2.145	1017
2028 Jul 15	0.75	94.03	94.06	unch.	2.017	1216
2029 Jan 15	0.875	93.3	94.01	unch.	2.06	1209
2029 Jan 15	2.5	102.01	102.04	unch.	2.077	1422
2029 Apr 15	3.875	109.03	109.06	unch.	2.125	1857
2029 Jul 15	0.25	90.16	90.19	unch.	1.964	1193
2030 Jan 15	0.125	88.21	88.24	-1	2.025	1186
2030 Jul 15	0.125	88.1	88.13	-2	1.943	1190
2031 Jan 15	0.125	87.06	87.08	-3	1.999	1172
2031 Jul 15	0.125	86.23	86.26	-2	1.947	1139
2032 Jan 15	0.125	85.22	85.25	-3	1.984	1101
2032 Apr 15	3.375	110.05	110.1	-4	2.058	1720
2032 Jul 15	0.625	89.04	89.09	-2	1.95	1050
2033 Jan 15	1.125	92.18	92.23	-4	1.984	1025
2033 Jul 15	1.375	94.28	95.02	-3	1.93	1005
2040 Feb 15	2.125	101.02	101.07	-12	2.037	1412
2041 Feb 15	2.125	101.05	101.1	-12	2.036	1394
2042 Feb 15	0.75	79.2	79.24	-12	2.077	1351
2043 Feb 15	0.625	76.2	76.24	-11	2.086	1328
2044 Feb 15	1.375	87.28	88.01	-13	2.099	1310
2045 Feb 15	0.75	76.22	76.27	-13	2.097	1296
2046 Feb 15	1	80.07	80.12	-14	2.103	1288
2047 Feb 15	0.875	77.09	77.14	-14	2.1	1264
2048 Feb 15	1	78.31	79.04	-14	2.097	1238
2049 Feb 15	1	78.14	78.2	-15	2.089	1213
2050 Feb 15	0.25	63.01	63.06	-14	2.065	1187
2051 Feb 15	0.125	59.26	59.31	-14	2.038	1172
2052 Feb 15	0.125	59.02	59.08	-14	2.014	1096
2053 Feb 15	1.5	87.26	88.09	-19	2.03	1027

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Binomial model of a two-period ZCB

A simple closed-form stochastic interest rate model

- Definitions
  - r_0 = spot rate for one period at time 0, known
  - r_1 = spot rate for one period at time 1, a random variable
  - r_{02} = spot rate for two-year debt
  - μ_r = expected change in spot rate from time 0 to 1
  - σ_r = standard deviation of change in spot rate from time 0 to 1
  - V_t = Value of the ZCB at time $t \in \{0, 1, 2\}$; $V_2 = 1$, V_1 is a random variable
  - k = cost of interest rate risk
  - f_1 = one-year forward rate at time 1

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This simple case can also be computed analytically

- Suppose X is a binomial random variable with two equally likely outcomes.
  - The two outcomes are: $X = a + b$ and $X = a - b$
  - Then $E[X] = a$ and $SD[X] = |b|$

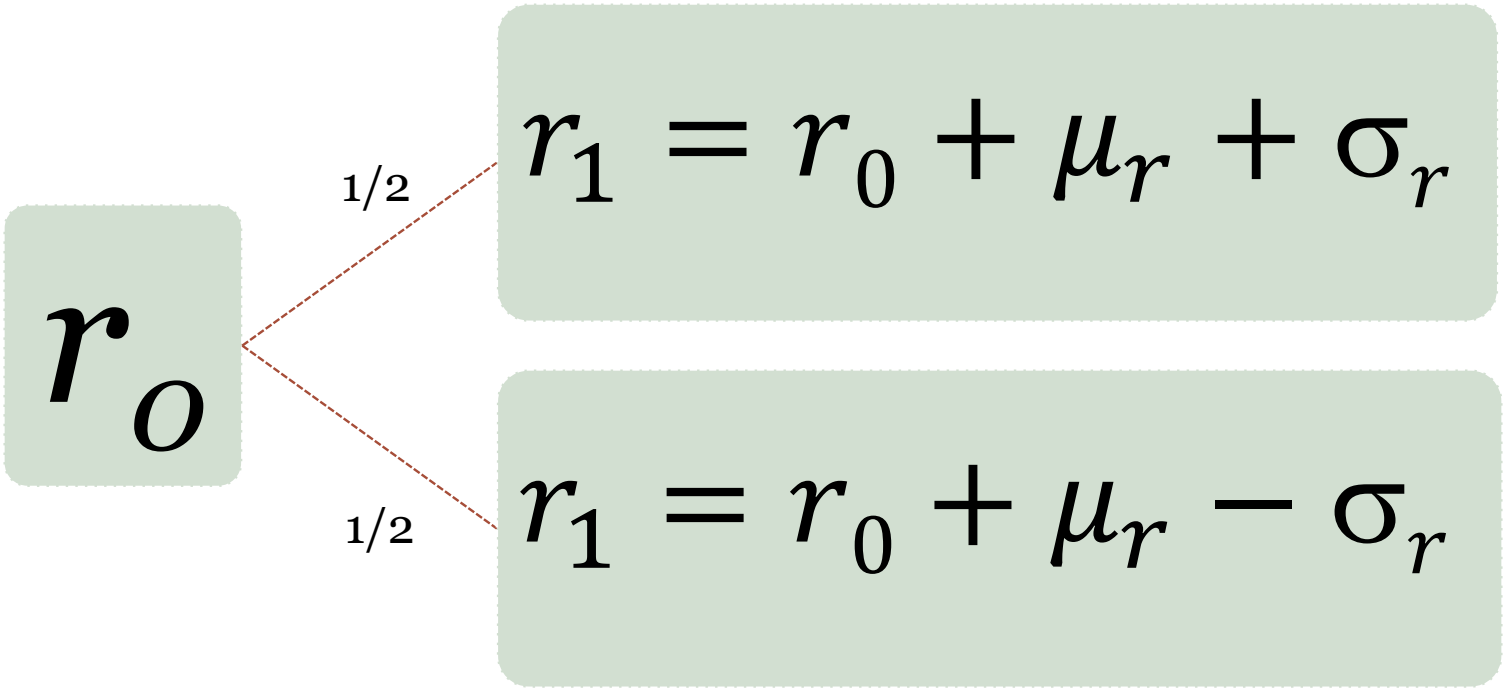
- Partial fractions: $\frac{1}{a+b} + \frac{1}{a-b} = \frac{2a}{a^2-b^2}$ $\frac{1}{a+b} - \frac{1}{a-b} = \frac{2b}{a^2-b^2}$

- Long division: $\frac{a^2-b^2}{a-kb} = a + kb - \frac{b^2(1-k^2)}{a-kb}$

Fun formulas!

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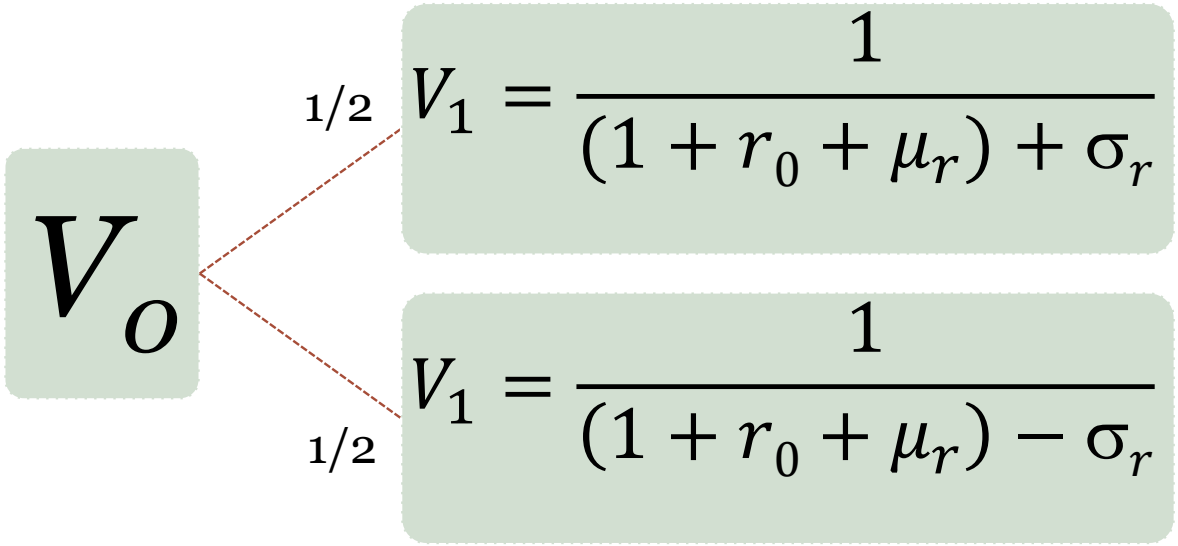
Short-term interest rate evolution



Hints: Let $a = (1 + r_0 + \mu_r)$ and $b = \sigma_r$; use discrete discounting

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Determining $V_1(r_1)$



Hint: Let $a = (1 + r_0 + \mu_r)$ and $b = \sigma_r$

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V_o with the GVE

- MBE
- $E[V_1] =$
- $SD[V_1] =$
- $V_0 =$

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Finding the spot and forward rate

- $V_0 = \frac{1}{(1+r_{02})^2} = \frac{1}{(1+r_0)(1+f_1)}$
- $(1 + f_1) = \frac{1}{(1+r_0)V_0} =$
- *Interpretations*

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Interpretations

- In this model, $f_1 \approx r_0 + \mu_r + k\sigma_r = E[r_1] + k\sigma_r$

- Forward prices rise approx. linearly with increasing spot rates, expected future spot rates, and the cost of interest rate risk
- Set some parameters equal to zero to isolate the effects of other parameters
- The pure convexity effect is found by setting k=0.
- The negative impact of convexity on the forward rate is negligible, on the order of σ_r^2

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Extensions of the MR TS model

- "Discrete-Time Continuous-State Interest Rate Models", Michael A. Sullivan, OCC Economics Working paper 2000-6
  - Multiperiod, plus option pricing (easiest to read)
- "One-factor interest rate models and the valuation of interest rate derivative securities", J. Hull and A. White, **Journal of Financial and Quantitative Analysis** 28, 235-254, 1993.
  - The basis for current market standards
- "An equilibrium characterization of the term structure", **Journal of Financial Economics** 5, 177-188.
  - First continuous time stochastic model (mean-reverting, with risk premium)
- "A continuous time approach to the pricing of bonds", Brennan, M. and Eduardo S. Schwartz. (1979).
  - The first two-factor model
- "Bond pricing and the term structure of interest rates.", Heath, Jarrow and Morton (1988).
  - Takes the initial term structure as given, and constructs arbitrage-free movements

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Today's Lecture

- We learned how to value riskless cash flows, even in the presence of stochastic (randomly changing) interest rates

  - Idealized bonds
  - Compounding conventions
  - GVE for bonds
  - Rates differ by period
  - Add risk, solve backwards (recursively)
  - Spot Forward and Expected spot rates
  - Term structure
  - US Government bonds: Bills, Notes, Bonds, STRIPs and TIPS
  - ZCBs bootstrapping
  - Implied parameters found in bond prices
    - Forward rates
    - Expected risk-adjusted inflation rates

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Review Questions

1. What are idealized bonds?
2. How is the effective annual yield of a bond calculated? How does it compare to a bond-equivalent yield?
3. What is the formula for the present value of a cash payment, discounted continuously?
4. How are ZCBs priced if the discount rate differs in every future year?
5. How are bonds priced (theoretically) in the presence of stochastic interest rates?
6. Explain spot rates, forward rates and expected future spot rates?
7. What is the general shape of the term structure? Why does it take that shape?
8. What might a downward sloping term structure suggest?
9. What is a forward bond?

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Review Questions

10. What is an FRN and how is it priced?
11. What is the present value of a payment that is proportional to a future interest rate?
12. What is mean by a factor model of the term structure?
13. What are the differences between bonds and loans?
14. What is a ZCB and why is it important?
15. Why do we see negative interest rates?
16. What happened to the composition of U.S. debt due to COVID?
17. Why is the yield on a T-Bill not really the yield?
18. What is a STRIP? What does it offer that differentiates it from T-Bills?
19. What is yield curve bootstrapping?

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Review Questions

21. Why are T-bonds preferred over STRIPs for bootstrapping?
22. How does accrued interest relate to bond pricing?
23. What is meant by the terms real and nominal?
24. How is the CPI computed?
25. What is a COLA, and how is present value computed in the presence of a COLA?
26. How can one determine implied inflation?

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